
VISION TRANSFORMERS ON FINE-GRAINED VISUAL CLASSIFICATION TASKS

DEEP LEARNING FOR VISUAL RECOGNITION - GROUP 20

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ABSTRACT

Fine-grained visual classification (FGVC) requires distinguishing subtle and localized differences between visually similar categories. In this project, we conducted a 9-class dog-breed classification task using three representative model families: a CNN baseline, a Vision Transformer (ViT), and a TransFG-inspired architecture (ViT-FG). The CNN and ViT baselines both achieve strong accuracy on the reduced dataset. The ViT-FG model, using a soft attention-weighted pooling mechanism, obtains the highest accuracy among the three models. Overall, our findings show that Vision Transformers perform at least as well as CNNs for small-scale FGVC tasks, and that attention-based part aggregation can further improve fine-grained recognition performance. ¹

1 Introduction

Image classification is one of the most fundamental tasks in computer vision, where the goal is to assign a label to an input image. While humans can easily recognize objects even under variations in pose, lighting, and background, machines have historically struggled with this task. From 2010 to 2017 the ImageNet Large Scale Visual Recognition Challenge [Russakovsky et al. 2015] was held annually to encourage development in this field, and oversaw the breakthrough of convolutional neural networks when AlexNet [Krizhevsky et al. 2012] won in 2012, and these kinds of models remained the state of the art for image classification tasks, until the development of the vision transformer. In an attempt to improve the performance of convolutional neural networks, an attention mechanism was developed, allowing models to focus on the most relevant regions of an input. The success of attention led to the development of the Transformer architecture — originally proposed for natural language processing — was later extended to computer vision through the Vision Transformer (ViT) [Dosovitskiy et al. 2020].

A subset of image classification is Fine-Grained Visual Classification (FGVC), where a model is tasked with distinguishing between members in the same category, like different species of flowers, different models of cars or different breeds of dogs. This is a harder task since the images will broadly be the same, and the model will have to learn to distinguish on the fine-grained details which differ.

In this project, we compare the performance of CNN and ViT on the task of Fine-Grained Visual Classification, our goal is to find out how these models differ in accuracy and interpretability, and to evaluate whether part-based attention aggregation can further improve ViT performance.

2 Related Work

Early approaches to Fine-Grained Visual Classification (FGVC) were primarily built on convolutional neural networks (CNNs).

¹The code can be found on the repository: <https://gitlab.au.dk>

However, CNNs are limited by local receptive fields, making it difficult to capture long-range dependencies essential for distinguishing highly similar object categories. The introduction of the Vision Transformer (ViT) [Dosovitskiy et al., 2020] marked a significant shift in image classification. By applying self-attention over non-overlapping patches, ViT can capture global relationships between distant regions, an ability particularly beneficial for FGVC tasks. The performance improvements brought by transformers are evident in benchmark studies. Table 1 summarizes results from the TransFG paper [He et al., 2022], comparing CNN-based methods and transformer-based methods on the Stanford Dogs dataset. ViT and TransFG outperform classical CNN approaches, reinforcing the view that global attention mechanisms are highly effective for FGVC tasks.

Method	Backbone	Top-1 Accuracy
MaxEnt	DenseNet-161	83.6%
FDL	DenseNet-161	84.9%
Cross-X	ResNet-50	88.9%
API-Net	ResNet-101	90.3%
ViT	ViT-B_16	91.7%
TransFG	ViT-B_16	92.3%

Table 1: FGVC benchmark results on the Stanford Dogs dataset

These results motivate the use of transformer-based architectures in our project, especially when dealing with fine-grained dog-breed recognition.

While TransFG [He et al., 2022] introduces a part-based mechanism that selects the top- k most important patch tokens according to the CLS-to-patch attention map, such hard patch selection has several limitations. First, the top- k operator is non-differentiable and may discard informative regions whose attention scores fall just below the threshold. Second, hard selection is less stable on smaller datasets, where attention distributions tend to fluctuate significantly during training.

Because our project runs on a subset of the Stanford Dogs dataset, stable gradient flow is particularly important. Therefore, we adopt a soft attention pooling strategy that aggregates all patch embeddings using normalized CLS-to-patch attention weights. This design choice is supported by prior works such as TokenLearner [Ryoo et al., 2021], which learns soft masks to reweight patch tokens, and CrossViT [Chen et al., 2021], which performs attention-weighted feature fusion across scales. These studies show that soft aggregation improves robustness, optimization smoothness, and generalization—properties well aligned with our fine-grained setting.

3 Methods

3.1 Dataset

For this task we have picked the Stanford Dogs Dataset [Khosla et al., 2011], a dataset built for training models for FGVC tasks, using images from ImageNet [Deng et al., 2009], and which was used as the dataset for a FGVC task in the ImageNet Large Scale Visual Recognition Challenge in 2012. It is difficult dataset to accurately classify, because of the similarities between different classes, and also the variation within classes as they all contain dogs of varying ages, colors and fur-length. The images are also taken of dogs from everyday life, meaning there are varying backgrounds and other objects that might be more prominent than the dog in the pictures. The dataset consists of 120 classes of different dog breeds with ~ 180 images of varying sizes in each class for a total of 20580 images in the dataset.

We picked out a subset of 9 classes of dogs from the dataset for faster development, rather than running experiments on all 120 classes, which can be prohibitively slow. The 9 picked classes were: Japanese spaniel, Shih-Tzu, Border collie, Maltese dog, Beagle, Kerry blue terrier, Irish terrier, Labrador retriever and German Shepherd (See Figure 1). The classes were picked because they are common dog breeds that would give a varied representation, but were also picked because of their similarities to each other that could challenge the model. The Japanese spaniel, Shih-Tzu and Maltese dog are all small white dogs that share similarities, the Kerry blue terrier and Irish terrier have very similar shapes, but distinct colors, the Labrador retriever has large color variation within the class, as they can be beige, brown or black and the Border collie has a similar color pattern to the Japanese spaniel. The expectation was therefore that if the model and parameters performed well on these classes, then it would also work well on the entire dataset.

This left us with a reduced dataset of 1667 images we could more easily train on. Since the images in the dataset are all of varying sizes, we squeeze all images down to be the same size of 224×224 pixels, with 3 channels for color. 80% of images in each class (so ~ 144 per class) are added to the training set with the remaining 20% (so ~ 36 per class) being used as a test set.

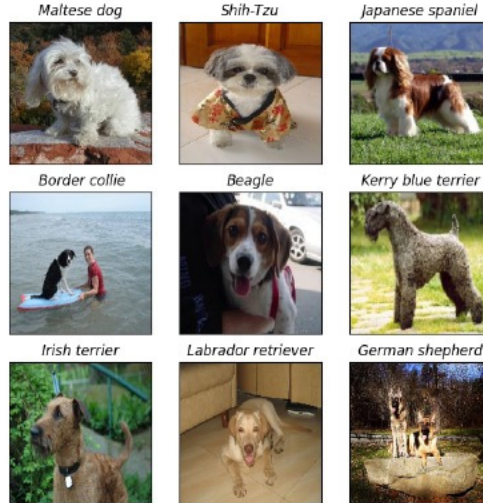


Figure 1: Sample images from the 9-class reduced dataset

3.2 Implementations

We have trained three different models. A Convolutional Neural Network, a baseline Vision Transformer and a Vision Transformer specialized for FGVC tasks. See the implementations [on the repository](#).

The CNN is built using ResNet50 as a pretrained encoder, and we then finetune the decoder on our dataset. This was done to hopefully be able to achieve good results on a difficult dataset, without spending too much time training a network from scratch. The decoder is built with a fully connected layer with a ReLU activation function and uses Kaiming weight initialization. We then have a dropout layer to try and reduce overfitting. Finally we have a fully connected layer with Softmax activation function that outputs the predictions of the model. This follows standard CNN fine-tuning practice, while keeping the backbone frozen to ensure stable training on a relatively small dataset.

As a second baseline, we employ a pretrained Vision Transformer (ViT-Small). We also attempted fine-tuning a ViT-Base model, however, the larger parameters caused severe overfitting on our relatively small dataset. Therefore, we adopted ViT-Small as a more stable and appropriately sized backbone model. It processes an image by dividing it into fixed-size patches, projecting each patch into a token embedding, and appending a learnable class token (CLS). These tokens are passed through multiple Transformer encoder layers composed of multi-head self-attention and feed-forward blocks. To adapt the pretrained model to our 9-class dog classification task, we freeze the entire transformer encoder (patch embedding, positional embeddings, and all attention blocks) and train only the final classification head. This approach treats the backbone as a fixed feature extractor, ensuring stable optimization and reducing the risk of overfitting given the limited amount of training data. The classification head is optimized using the AdamW optimizer with a small learning rate.

Our ViT-FG model is inspired by TransFG [He et al. \[2022\]](#), but replaces its hard top- k patch selection with a soft attention pooling mechanism. For each image, we extract the CLS-to-patch attention map from the last Transformer block using a lightweight AttentionSpy wrapper, which records attention matrices during forward propagation without modifying the underlying computation. The CLS-to-patch attention vector is normalized with a softmax operation and interpreted as continuous importance weights over all patch tokens. These weights are then used to compute a weighted sum of patch embeddings, producing a fine-grained “part feature” that remains fully differentiable. The resulting part feature is concatenated with the global CLS token, followed by layer normalization and a linear classifier. In contrast to hard top- k selection, soft attention pooling retains all spatial information, avoids discontinuous gradients, and provides more stable training on small datasets.

3.3 Data Augmentation

We use different augmentation strategies for the CNN and ViT models, tailored to their architectural characteristics.

For CNN, we utilize its inherent translation invariance by applying random rotations ($\pm 20^\circ$), horizontal flips, and spatial shifts (up to 10%) via ImageDataGenerator. These geometric perturbations enhance robustness to viewpoint variations without compromising the local textures critical for fine-grained classification. These augmentations increase

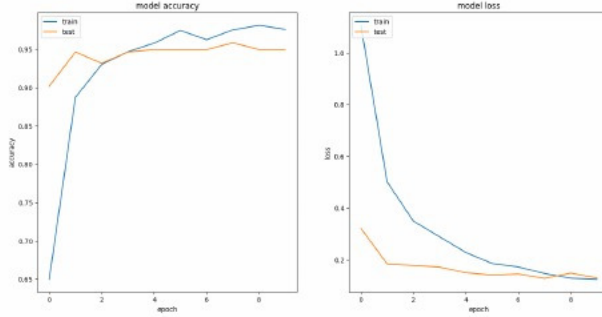
robustness to positional and viewpoint variations without significantly distorting the fine-grained patterns essential for dog-breed classification.

In contrast to the CNN, our ViT augmentation avoid strong affine transformations (e.g., shear, large rotation, vertical flips), ViTs are more affected by spatial distortions due to their fixed patch-tokenization process. We therefore focus on augmentations that preserve spatial structure, ensuring inputs align with the pretrained feature extractor while providing sufficient variability for effective fine-tuning. Finally, images are normalized using ImageNet mean and standard deviation to ensure compatibility with the pretrained ViT backbone.

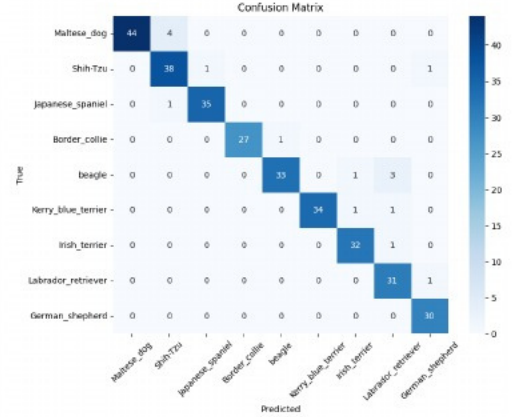
4 Results

All models were trained for 10–15 epochs using a batch size of 32. We present the resulting accuracy and loss curves for each model, along with confusion matrices that summarize their prediction behavior on the test set.

Figure 2a shows the accuracy and loss curves for the CNN, where we can see the accuracy on the test set improves slightly for the first three epochs, hitting around 95%, but remains mostly flat afterwards, with the model not really improving afterwards. Figure 2b shows the confusion matrix with most predictions lying correctly on the diagonal, with a few Maltese dogs being classified as Shi-Tzus, some Beagles as Labrador Retriever and some other single outliers, but not any large groups all being mislabeled.

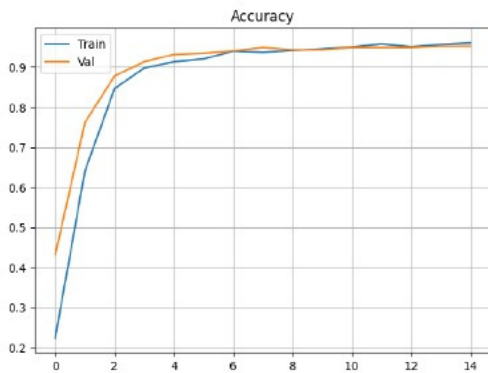


(a) Training curves CNN

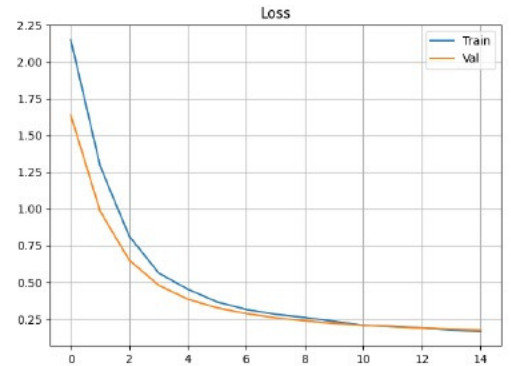


(b) Confusion Matrix for CNN

The results for the visions transformers are quite similar, with the accuracies of the ViT in Figure 3a and ViT-FG in Figure 4a plateauing quite quickly. The confusion matrices in Figure 5 show the predictions are mostly correct, with some slight confusion between Maltese dogs and Shi-Tzuz.

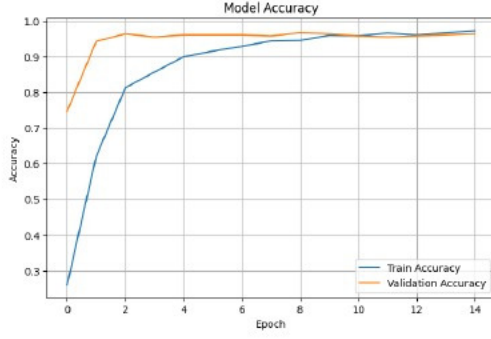


(a) Accuracy

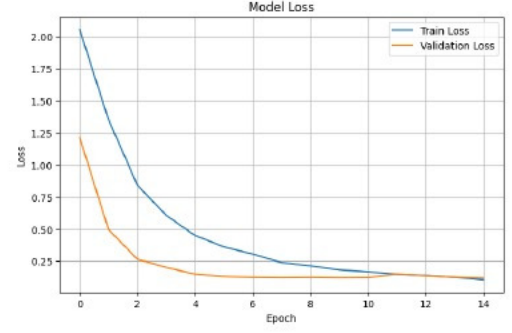


(b) Loss

Figure 3: Training curves of ViT

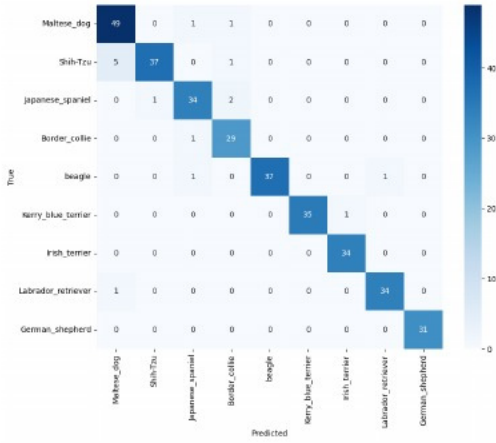


(a) Accuracy

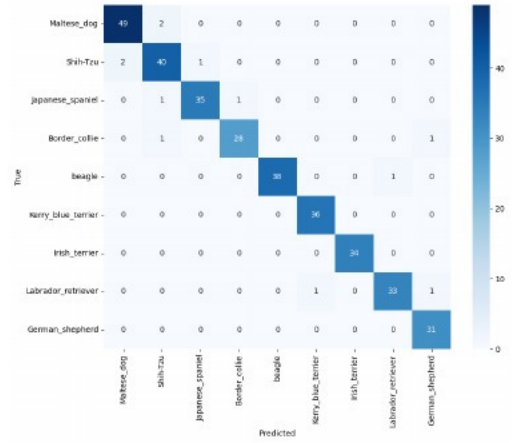


(b) Loss

Figure 4: Training curves of ViT-FG



(a) Confusion Matrix for ViT



(b) Confusion Matrix for ViT-FG

Figure 5: Comparison of Confusion Matrices for ViT and ViT-FG

To better understand how the transformer-based model makes fine-grained predictions, we evaluate the interpretability of our ViT-FG architecture using **Attention Rollout**. We extract the attention matrices from all Transformer blocks and compute the rollout, producing a global attention map that highlights salient regions in the image.

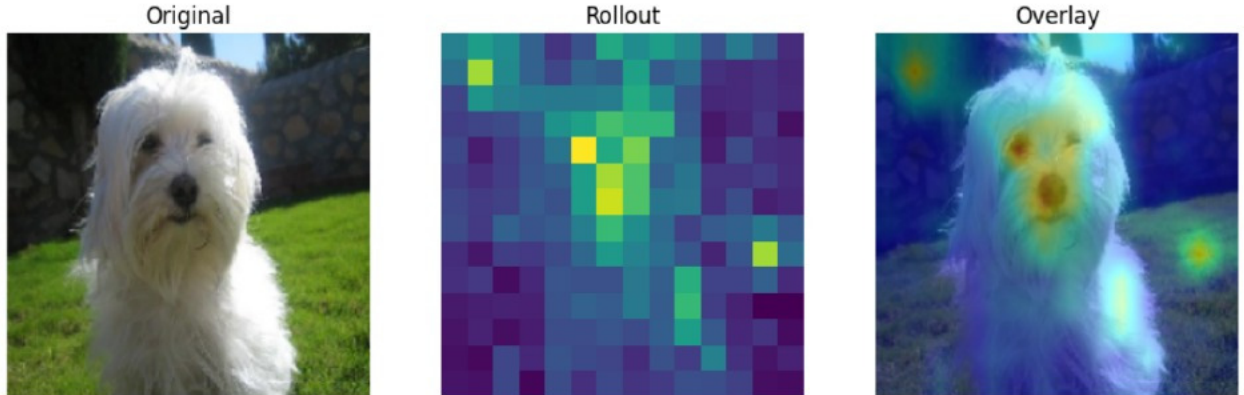


Figure 6: Attention Rollout visualization for ViT-FG. From left to right: the original image, the accumulated rollout attention heatmap, and the overlay.

The rollout heatmap correctly focuses on the dog’s face—particularly the eyes, nose, and characteristic fur texture. This demonstrates that our soft attention pooling mechanism successfully captures subtle discriminative cues rather than only relying on background correlations.

The models performed quite well on classifying the reduced dataset, all achieving more than 95% accuracy on the test set, with the CNN and the base ViT achieving very similar results, and the specialized ViT-FG performing slightly better.

Model	Accuracy
CNN	95.5%
ViT_base	95.2%
ViT-FG	96.7%

Table 2: Models evaluated on the 9-class reduced dataset

We also attempted to train the models on the entire Stanford Dogs dataset, and after a little more than 2 hours training the CNN was up to 63.2% accuracy. The vision transformers however, were much slower when run on all 20,000 images and we were unable to train them for 10 epochs. We ran the baseline ViT for only 1 epoch, which took around 1.5 hours, but even then already got an accuracy of 67.6%, beating the CNN.

5 Discussion

We did not perform any hyperparameter search, instead experimenting and adjusting parameters by hand. This would probably be the next step to look into if we wanted to increase the performance of our models more.

We also face computational constraints when evaluating the vision transformers on the entire Stanford Dogs dataset, but with some optimizations and better hardware, we would expect that the trend of the ViT-FG outperforming the other models continues, especially as vision transformers should perform better on large datasets.

Another limitation is that the three models are not trained under identical optimization conditions. The CNN and the baseline ViT both freeze their feature extractors and train only the classification head, whereas the ViT-FG model performs full finetuning of the transformer encoder with a small learning rate. This difference means that part of the performance gain observed in ViT-FG may come from updating more parameters rather than solely from the proposed soft attention pooling mechanism.

We also observe that all models converge very quickly on the reduced 9-class subset, reaching near-optimal accuracy within only a few epochs. This suggests that the task is comparatively easy for modern pretrained feature extractors, which may explain why the absolute accuracy differences between the three models remain small.

Finally, preliminary experiments on the full Stanford Dogs dataset show that the ViT already surpasses the CNN after only a single epoch, indicating that transformer-based architectures scale more favorably with dataset size.

6 Conclusion

In this project, we evaluated a CNN, a baseline Vision Transformer, and a ViT-FG model on a 9-class fine-grained dog-breed classification task. All three models reached strong accuracy, showing that modern pretrained ViTs perform at least as well as CNNs for fine-grained recognition. Among them, the ViT-FG model achieved the best results, suggesting that adding a simple part-aware attention aggregation can provide measurable improvements over a standard ViT.

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